

EEEE- 414: Classical Control

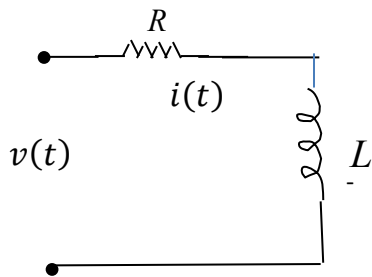
Homework Assignment # 1

Due date: by 11:59 PM, Thursday, September 3, 2026

Problem #1

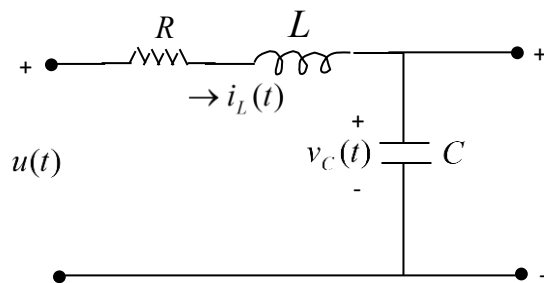
Given the circuit shown below:

- Write the differential equation for the circuit if $v(t) = u(t)$, a unit step.
- Solve the differential equation for the current $i(t)$, if there is no initial energy in the circuit.
- Make a plot of your solution (use MATLAB) if $R/L = 1$.



Problem # 2

Repeat problem #1 for the following RLC circuit. Assume $R = 1\Omega$, $L = 0.5\text{H}$, and $C = \frac{1}{8}\text{F}$



Problem # 3

Assuming zero initial conditions, find solutions for the following differential equations.

- $\frac{dx(t)}{dt} + 5x(t) = 2 \cos(3t)$
- $\frac{d^2x(t)}{dt^2} + 4 \frac{dx(t)}{dt} + 2x(t) = 2\sin(t)$
- $\frac{d^2x(t)}{dt^2} + 6 \frac{dx(t)}{dt} + 20x(t) = 5u(t)$

Problem # 4

Solve the following differential equations:

a) $\frac{d^2x(t)}{dt^2} + 2\frac{dx(t)}{dt} + 2x(t) = \sin(2t)$

$$x(0) = 2$$

$$\frac{dx(t)}{dt}(t=0) = -3$$

b) $\frac{d^2x(t)}{dt^2} + 2\frac{dx(t)}{dt} + x(t) = 5e^{-2t} + t$

$$x(0) = 2$$

$$\frac{dx(t)}{dt}(t=0) = 1$$

c) $\frac{d^2x(t)}{dt^2} + 4x(t) = t^2$

$$x(0) = 1$$

$$\frac{dx(t)}{dt}(t=0) = 2$$

Problem # 5

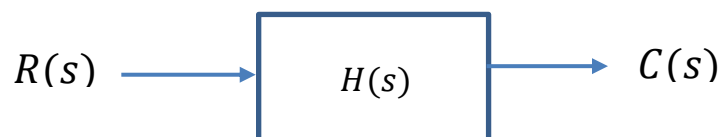
A system is described by the following differential equation:

$$\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + y(t) = \frac{d^3x(t)}{dt^3} + 4\frac{d^2x(t)}{dt^2} + 6\frac{dx(t)}{dt} + 8x(t)$$

Find the expression for the transfer function of the systems $H(s) = \frac{Y(s)}{X(s)}$

Problem # 6

Write the differential equation for the system shown below:



$$\frac{C(s)}{R(s)} = \frac{s^5 + 2s^4 + 4s^3 + 4s^2 + 4}{s^6 + 7s^5 + 3s^4 + 2s^3 + s^2 + 5}$$

Problem # 7

Use MATLAB to generate the partial fraction expansion of the following transfer function.

$$F(s) = \frac{10^4(s+5)(s+70)}{s(s+45)(s+55)(s^2+7s+110)(s^2+6s+95)}$$

Problem # 8

Find the transfer function, $G(s) = \frac{V_0(s)}{V_1(s)}$ for each operational amplifier circuit shown below.

